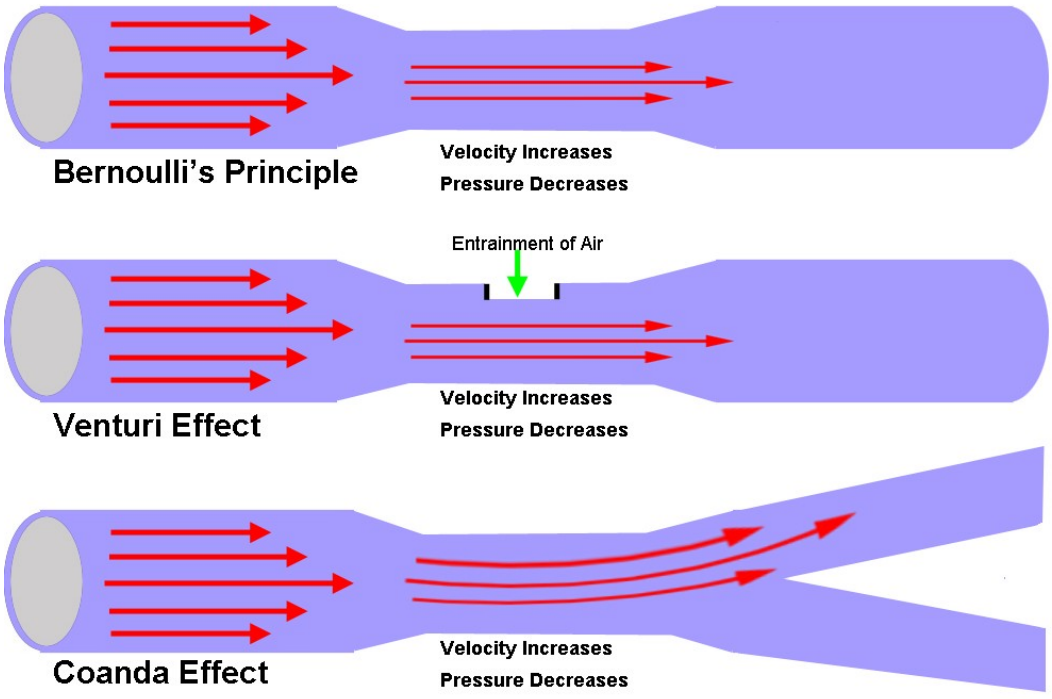
Bernoulli's principle states that the flow of a gas or fluid through a tube increases if the diameter of the tube narrows. At the same time, there is a drop in pressure in the area of the constriction. For example, if oxygen flows through an anesthesia circuit at 2 L/min, and you attach an endotracheal tube through it that has half the cross-sectional area of the anesthesia circuit, the flow rate of oxygen at the end of the endotracheal tube would be 4L/min.

The Venturi effect utilizes the Bernoulli principle to entrain gas into a tube. As the Bernoulli principle describes, there is a decrease in pressure at the point of narrowing in a tube. If you place an opening in the tube at that point, the decreased pressure will act as a vacuum and draw in gas through the opening. This principle is utilized by both jet ventilation techniques and nebulizers that can deliver medication into a breathing circuit.

The Coanda effect describes the predisposition for a fluid or gas to follow a curved path at the bifurcation point of a tube. As the cross-sectional area of the tubes increases, the velocity of the gas or fluid decreases and the pressure increases. The tube in which the pressure increases last tends to receive a greater distribution of the gas or fluid. This effect can be of consequence in blood vessels or in the pulmonary airways where blood or oxygen preferentially flow to one area over another.



Nagelhout JJ, Plaus K. Nurse Anesthesia. 5th ed. St. Louis, MO: Elsevier Saunders Company; 2014: 226-227.

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